

Chapter 8: Complex Vector Spaces

Chapter overview

Complex numbers extend real arithmetic so that every polynomial equation can be treated within one number system. The chapter develops complex arithmetic, polar geometry, complex vector spaces, complex inner products, unitary matrices, Hermitian matrices, and the complex spectral theorem.

Original contexts

All worked numerical examples are original. Themes selectively use the San Antonio Spurs, Backstreet Boys, hiking, badminton, driving, travel, and R&B music. Signal and data interpretations are fictional teaching models.

Inner-product convention

Throughout these notes,

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{j=1}^n u_j \overline{v_j}.$$

This convention is linear in the first argument and conjugate-linear in the second.

1 Complex Numbers

Complex number

A complex number is

$$z = a + bi, \quad a, b \in \mathbb{R}, \quad i^2 = -1.$$

The real part is $\operatorname{Re}(z) = a$ and the imaginary part is $\operatorname{Im}(z) = b$.

The standard form is unique. If $a + bi = c + di$, then

$$(a - c) + (b - d)i = 0.$$

Because 1 and i are linearly independent over $\mathbb{R}$, $a = c$ and $b = d$.

The complex plane

The number $a + bi$ is represented by the point (a, b) or the vector from the origin to (a, b) . The real axis is horizontal and the imaginary axis is vertical.

Detailed arithmetic

Let

$$z = 4 + 3i, \quad w = -2 + 5i.$$

Then

$$\begin{aligned} z + w &= (4 - 2) + (3 + 5)i = 2 + 8i, \\ z - w &= (4 + 2) + (3 - 5)i = 6 - 2i, \end{aligned}$$

and

$$\begin{aligned}zw &= (4 + 3i)(-2 + 5i) \\&= -8 + 20i - 6i + 15i^2 \\&= -23 + 14i.\end{aligned}$$

Multiplication formula

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

Proof

Expand by distributivity:

$$ac + adi + bci + bdi^2 = ac - bd + (ad + bc)i.$$

Quadratic equations

For

$$p(x) = x^2 - 6x + 13,$$

the discriminant is

$$(-6)^2 - 4(1)(13) = -16.$$

Thus

$$x = \frac{6 \pm 4i}{2} = 3 \pm 2i.$$

Conjugate-pair property

If a polynomial has real coefficients and z is a root, then $\bar{z}$ is also a root.

Proof

Let $p(x) = \sum_{j=0}^n a_j x^j$ with $a_j \in \mathbb{R}$. If $p(z) = 0$, then

$$p(\bar{z}) = \sum_{j=0}^n a_j \bar{z}^j = \overline{\sum_{j=0}^n a_j z^j} = \overline{p(z)} = 0.$$

Complex matrices

Let

$$A = \begin{bmatrix} 1+i & 2 \\ 3-i & -i \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1-i \\ i & 3 \end{bmatrix}.$$

Then

$$A + B = \begin{bmatrix} 3+i & 3-i \\ 3 & 3-i \end{bmatrix}.$$

For the product,

$$AB = \begin{bmatrix} 2+4i & 8 \\ 7-2i & 2-7i \end{bmatrix}.$$

The determinant is

$$\begin{aligned}\det(A) &= (1+i)(-i) - 2(3-i) \\ &= 1 - i - 6 + 2i \\ &= -5 + i \neq 0.\end{aligned}$$

Section 8.1 summary

Complex arithmetic follows ordinary algebra together with $i^2 = -1$. In the complex plane, addition is vector addition. Polynomial roots with real coefficients occur in conjugate pairs.

2 Conjugates, Modulus, and Division

Complex conjugate

For $z = a + bi$,

$$\bar{z} = a - bi.$$

Geometrically, conjugation reflects z across the real axis.

Conjugate identities

$$\overline{z + w} = \bar{z} + \bar{w}, \quad \overline{zw} = \bar{z}\bar{w}, \quad \overline{\bar{z}} = z.$$

If $w \neq 0$,

$$\overline{\left(\frac{z}{w}\right)} = \frac{\bar{z}}{\bar{w}}.$$

Proof of the product identity

Let $z = a + bi$ and $w = c + di$. Since

$$zw = (ac - bd) + (ad + bc)i,$$

$$\overline{zw} = (ac - bd) - (ad + bc)i.$$

Also,

$$(a - bi)(c - di) = (ac - bd) - (ad + bc)i.$$

Hence $\overline{zw} = \bar{z}\bar{w}$.

Modulus

$$|a + bi| = \sqrt{a^2 + b^2}.$$

It is the distance from the origin to (a, b) .

The conjugate satisfies

$$\begin{aligned} z\bar{z} &= (a + bi)(a - bi) \\ &= a^2 + b^2 \\ &= |z|^2. \end{aligned}$$

Product and quotient rules

$$|zw| = |z||w|, \quad \left|\frac{z}{w}\right| = \frac{|z|}{|w|} \quad (w \neq 0).$$

Proof

$$|zw|^2 = (zw)\overline{zw} = (z\bar{z})(w\bar{w}) = |z|^2|w|^2.$$

Both sides are nonnegative, so $|zw| = |z||w|$. Applying this to $(z/w)w = z$ gives the quotient rule.

Division

For $w = c + di \neq 0$,

$$\frac{a + bi}{c + di} = \frac{(a + bi)(c - di)}{c^2 + d^2}.$$

For example,

$$\begin{aligned}\frac{5 + 7i}{2 - i} &= \frac{(5 + 7i)(2 + i)}{5} \\ &= \frac{3 + 19i}{5} \\ &= \frac{3}{5} + \frac{19}{5}i.\end{aligned}$$

Inverse of a complex matrix

Let

$$A = \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix}.$$

Then

$$\det(A) = 1 - i^2 = 2,$$

so

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix}.$$

Verification gives

$$AA^{-1} = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = I.$$

Section 8.2 summary

Conjugation turns complex denominators into positive real numbers and converts modulus into algebra through $z\bar{z} = |z|^2$.

3 Polar Form and De Moivre's Theorem

Polar form

For $z = a + bi \neq 0$,

$$z = r(\cos \theta + i \sin \theta) = r \operatorname{cis} \theta,$$

where

$$r = |z|, \quad a = r \cos \theta, \quad b = r \sin \theta.$$

Arguments differ by integer multiples of 2π . The principal argument satisfies

$$-\pi < \operatorname{Arg}(z) \leq \pi.$$

Rectangular to polar form

For $z = -\sqrt{3} + i$,

$$r = \sqrt{3 + 1} = 2.$$

The point lies in Quadrant II and has reference angle $\pi/6$, so

$$\operatorname{Arg}(z) = \frac{5\pi}{6}.$$

Therefore,

$$z = 2 \operatorname{cis} \left(\frac{5\pi}{6} \right).$$

Product and quotient in polar form

If $z_1 = r_1 \operatorname{cis} \theta_1$ and $z_2 = r_2 \operatorname{cis} \theta_2$, then

$$z_1 z_2 = r_1 r_2 \operatorname{cis}(\theta_1 + \theta_2),$$

and, for $z_2 \neq 0$,

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \operatorname{cis}(\theta_1 - \theta_2).$$

Proof of the product formula

Expand and use the angle-addition identities:

$$\begin{aligned} z_1 z_2 &= r_1 r_2 [(\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) \\ &\quad + i(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2)] \\ &= r_1 r_2 \operatorname{cis}(\theta_1 + \theta_2). \end{aligned}$$

De Moivre's theorem

For every positive integer n ,

$$(r \operatorname{cis} \theta)^n = r^n \operatorname{cis}(n\theta).$$

Proof by induction

The statement is immediate for $n = 1$. Assume

$$(r \operatorname{cis} \theta)^n = r^n \operatorname{cis}(n\theta).$$

Then

$$\begin{aligned}(r \operatorname{cis} \theta)^{n+1} &= (r \operatorname{cis} \theta)^n (r \operatorname{cis} \theta) \\ &= r^n \operatorname{cis}(n\theta) r \operatorname{cis} \theta \\ &= r^{n+1} \operatorname{cis}((n+1)\theta).\end{aligned}$$

For example,

$$1 + i = \sqrt{2} \operatorname{cis}\left(\frac{\pi}{4}\right),$$

so

$$(1 + i)^8 = (\sqrt{2})^8 \operatorname{cis}(2\pi) = 16.$$

n th roots

The n distinct roots of $z = r \operatorname{cis} \theta$ are

$$w_k = r^{1/n} \operatorname{cis}\left(\frac{\theta + 2\pi k}{n}\right), \quad k = 0, \dots, n-1.$$

Derivation

Write $w = \rho \operatorname{cis} \phi$. The equation $w^n = z$ implies

$$\rho^n = r \quad \text{and} \quad n\phi = \theta + 2\pi k.$$

Therefore,

$$\rho = r^{1/n}, \quad \phi = \frac{\theta + 2\pi k}{n}.$$

The values $k = 0, \dots, n-1$ are distinct modulo 2π and every other integer repeats one of them.

The fourth roots of 16 are

$$2, \quad 2i, \quad -2, \quad -2i.$$

Section 8.3 summary

Polar form separates magnitude and direction. Multiplication adds angles, powers multiply angles, and roots divide angles into equally spaced directions.

4 Complex Vector Spaces and Inner Products

A complex vector space uses complex scalars. Thus

$$c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k$$

allows $c_j \in \mathbb{C}$.

A basis for $\mathbb{C}^3$

Let

$$\mathbf{v}_1 = (i, 0, 0), \quad \mathbf{v}_2 = (i, i, 0), \quad \mathbf{v}_3 = (0, 0, i).$$

If

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 = \mathbf{0},$$

then

$$(c_1 + c_2)i = 0, \quad c_2i = 0, \quad c_3i = 0.$$

Hence $c_1 = c_2 = c_3 = 0$. The set is independent and, because it contains three vectors in $\mathbb{C}^3$, it is a basis.

Complex Euclidean inner product

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{j=1}^n u_j \overline{v_j}.$$

Conjugation is essential. Without it,

$$(1, i) \cdot (1, i) = 1 + i^2 = 0,$$

although $(1, i) \neq \mathbf{0}$. With conjugation,

$$\langle (1, i), (1, i) \rangle = 1 + i(-i) = 2.$$

Inner-product axioms

$$\langle \mathbf{u}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{u} \rangle},$$

$$\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle,$$

$$\langle \alpha \mathbf{u}, \mathbf{v} \rangle = \alpha \langle \mathbf{u}, \mathbf{v} \rangle,$$

and

$$\langle \mathbf{u}, \mathbf{u} \rangle \geq 0,$$

with equality only for $\mathbf{u} = \mathbf{0}$.

Positive definiteness

$$\langle \mathbf{u}, \mathbf{u} \rangle = \sum_{j=1}^n u_j \overline{u_j} = \sum_{j=1}^n |u_j|^2.$$

Each term is nonnegative, and the sum is zero only when every component is zero.

Norm, distance, and orthogonality

$$\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}, \quad d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$

Vectors are orthogonal when $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

Detailed inner product

Let

$$\mathbf{u} = (2 + i, 1 - i, -1 + 2i), \quad \mathbf{v} = (1 - i, 2, 3 + i).$$

Then

$$\begin{aligned} \langle \mathbf{u}, \mathbf{v} \rangle &= (2 + i)(1 + i) + 2(1 - i) + (-1 + 2i)(3 - i) \\ &= 2 + 8i. \end{aligned}$$

Cauchy–Schwarz inequality

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|.$$

Proof

Assume $\mathbf{v} \neq \mathbf{0}$ and set

$$\alpha = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle}.$$

Then

$$0 \leq \|\mathbf{u} - \alpha \mathbf{v}\|^2 = \|\mathbf{u}\|^2 - \frac{|\langle \mathbf{u}, \mathbf{v} \rangle|^2}{\|\mathbf{v}\|^2}.$$

Multiplying by $\|\mathbf{v}\|^2$ gives the squared inequality. Taking nonnegative square roots completes the proof.

Projection

For $\mathbf{v} \neq \mathbf{0}$,

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle} \mathbf{v}.$$

The residual is orthogonal:

$$\langle \mathbf{u} - \text{proj}_{\mathbf{v}} \mathbf{u}, \mathbf{v} \rangle = 0.$$

Complex Gram–Schmidt

$$\mathbf{w}_1 = \mathbf{v}_1, \quad \mathbf{w}_2 = \mathbf{v}_2 - \frac{\langle \mathbf{v}_2, \mathbf{w}_1 \rangle}{\langle \mathbf{w}_1, \mathbf{w}_1 \rangle} \mathbf{w}_1.$$

The formulas are identical in structure to the real case, but every projection coefficient uses the complex inner product.

Section 8.4 summary

Conjugation produces a positive real squared norm and changes the real dot product into a valid complex inner product. Projection and Gram–Schmidt then extend naturally.

5 Unitary and Hermitian Matrices

Conjugate transpose

$$A^* = \overline{A}^T.$$

Equivalently,

$$(A^*)_{ij} = \overline{a_{ji}}.$$

Conjugate-transpose rules

$$\begin{aligned} (A + B)^* &= A^* + B^*, & (cA)^* &= \bar{c} A^*, \\ (AB)^* &= B^* A^*, & (A^*)^* &= A. \end{aligned}$$

Proof that $(AB)^* = B^* A^*$

The (i, j) entry of $(AB)^*$ is

$$\overline{(AB)_{ji}} = \sum_k \overline{a_{jk} b_{ki}}.$$

The (i, j) entry of $B^* A^*$ is

$$\sum_k \overline{b_{ki} a_{jk}}.$$

The entries are equal because complex multiplication is commutative.

Unitary matrix

A square complex matrix U is unitary when

$$U^* U = U U^* = I.$$

Thus $U^{-1} = U^*$.

The columns of U form an orthonormal basis exactly when $U^* U = I$.

Unitary audio mixer

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix}, \quad U^* = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix}.$$

Then

$$U^* U = I.$$

Unitary matrices preserve geometry

$$\langle U\mathbf{x}, U\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle,$$

so

$$\|U\mathbf{x}\| = \|\mathbf{x}\|.$$

Proof

Using the conjugate-transpose representation of the inner product,

$$\langle U\mathbf{x}, U\mathbf{y} \rangle = \mathbf{x}^T U^T \overline{U} \mathbf{y}.$$

The unitary relation implies $U^T \overline{U} = I$, so the expression equals $\mathbf{x}^T \mathbf{y}$.

Hermitian matrix

A square complex matrix H is Hermitian when

$$H^* = H.$$

Its diagonal entries are real and $h_{ji} = \overline{h_{ij}}$.

Hermitian eigenvalues are real

Every eigenvalue of a Hermitian matrix is real.

Proof

Let $H\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq 0$. Then

$$\langle H\mathbf{v}, \mathbf{v} \rangle = \lambda \langle \mathbf{v}, \mathbf{v} \rangle.$$

Since $H = H^*$,

$$\langle H\mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{v}, H\mathbf{v} \rangle = \overline{\lambda} \langle \mathbf{v}, \mathbf{v} \rangle.$$

Because $\langle \mathbf{v}, \mathbf{v} \rangle > 0$, $\lambda = \overline{\lambda}$, so λ is real.

Distinct Hermitian eigenspaces are orthogonal

If $H\mathbf{u} = \lambda\mathbf{u}$ and $H\mathbf{v} = \mu\mathbf{v}$ with $\lambda \neq \mu$, then

$$\langle \mathbf{u}, \mathbf{v} \rangle = 0.$$

Proof

$$\lambda \langle \mathbf{u}, \mathbf{v} \rangle = \langle H\mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, H\mathbf{v} \rangle = \mu \langle \mathbf{u}, \mathbf{v} \rangle.$$

Hence $(\lambda - \mu) \langle \mathbf{u}, \mathbf{v} \rangle = 0$. Since $\lambda \neq \mu$, the inner product is zero.

Complete unitary diagonalization

Let

$$H = \begin{bmatrix} 2 & i \\ -i & 2 \end{bmatrix}.$$

Its characteristic polynomial is

$$\begin{aligned} \det(H - \lambda I) &= (2 - \lambda)^2 - i(-i) \\ &= (2 - \lambda)^2 - 1 \\ &= (\lambda - 1)(\lambda - 3). \end{aligned}$$

For $\lambda = 1$, an eigenvector is $(-i, 1)^T$. For $\lambda = 3$, an eigenvector is $(i, 1)^T$. After normalization,

$$\mathbf{p}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} -i \\ 1 \end{bmatrix}, \quad \mathbf{p}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} i \\ 1 \end{bmatrix}.$$

Set

$$P = \frac{1}{\sqrt{2}} \begin{bmatrix} -i & i \\ 1 & 1 \end{bmatrix}.$$

Then

$$P^*P = I, \quad P^*HP = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}.$$

Complex spectral theorem

Every Hermitian matrix has an orthonormal basis of eigenvectors and is unitarily diagonalizable:

$$H = PDP^*,$$

where P is unitary and D is real diagonal.

Proof by induction

For $n = 1$, the result is immediate. Assume the theorem holds for Hermitian matrices of order $n - 1$, and let H be Hermitian of order n .

The characteristic polynomial has a complex root, so H has an eigenvalue λ_1 and a nonzero eigenvector $\mathbf{v}_1$. The eigenvalue is real by the preceding theorem. Normalize $\mathbf{v}_1$.

Let

$$W = \mathbf{v}_1^\perp.$$

If $\mathbf{w} \in W$, then

$$\begin{aligned} \langle H\mathbf{w}, \mathbf{v}_1 \rangle &= \langle \mathbf{w}, H\mathbf{v}_1 \rangle \\ &= \langle \mathbf{w}, \lambda_1 \mathbf{v}_1 \rangle \\ &= \overline{\lambda_1} \langle \mathbf{w}, \mathbf{v}_1 \rangle \\ &= 0. \end{aligned}$$

Thus W is invariant under H . The restriction of H to W is Hermitian. By the induction hypothesis, W has an orthonormal eigenbasis $\mathbf{v}_2, \dots, \mathbf{v}_n$. Together with $\mathbf{v}_1$, these vectors form an orthonormal eigenbasis of $\mathbb{C}^n$. Placing them in the columns of P gives $P^*HP = D$.

Section 8.5 summary

Unitary matrices are the complex analogues of orthogonal matrices. Hermitian matrices are the complex analogues of real symmetric matrices, with real eigenvalues and orthonormal eigenbases.

Terminology tables

Section 8.1

Term	Meaning
Complex number	$a + bi$ with $a, b \in \mathbb{R}$
Complex plane	Plane with real and imaginary axes
Complex matrix	Matrix with entries in $\mathbb{C}$
Conjugate pair	$a + bi$ and $a - bi$

Section 8.2

Term	Meaning
Conjugate	$\overline{a + bi} = a - bi$
Modulus	$ a + bi = \sqrt{a^2 + b^2}$
Rationalization	Multiplication by a conjugate to obtain a real denominator

Section 8.3

Term	Meaning
Argument	Angle locating a complex number
Principal argument	Argument in $(-\pi, \pi]$
Polar form	$r \operatorname{cis} \theta$
De Moivre's theorem	$(r \operatorname{cis} \theta)^n = r^n \operatorname{cis}(n\theta)$

Sections 8.4–8.5

Term	Meaning
Complex inner product	Positive-definite sesquilinear form
Conjugate transpose	$A^* = \overline{A}^T$
Unitary matrix	$U^*U = I$
Hermitian matrix	$H^* = H$
Unitary diagonalization	$P^*HP = D$

Chapter closing

Complex vector spaces preserve the structure of real linear algebra while adding conjugation. The conjugate is the mechanism that makes magnitude, inner products, adjoints, and spectral theory work correctly over $\mathbb{C}$.